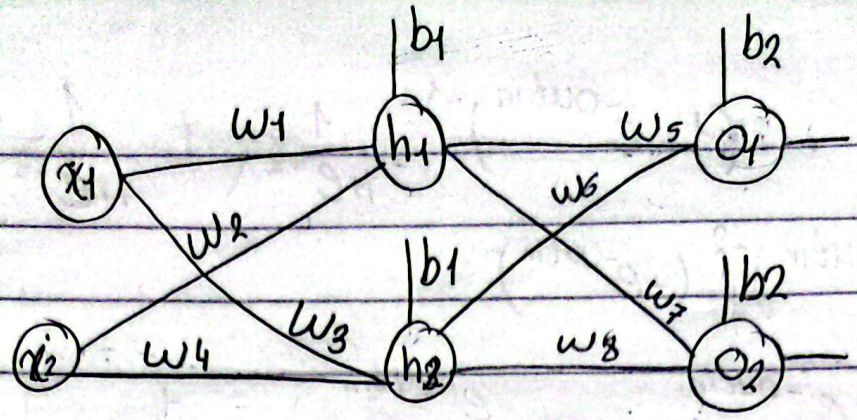

x_1	x_2	desired-1	desired-2
0.05	0.10	0.01	0.99

w_1	w_2	w_3	w_4	w_5	w_6	w_7	w_8	b_1	b_2
0.15	0.2	0.25	0.3	0.4	0.45	0.5	0.55	0.35	0.6

1) $net_h_1 = w_1 x_1 + w_2 x_2 + b_1$

$= 0.15 \times 0.05 + 0.2 \times 0.1 + 0.35 = 0.3775$

2) $out_h_1 = \frac{1}{1+e^{-h_1}} \rightarrow \frac{1}{1+e^{-0.3775}} = 0.5933$

3) $net_h_2 = w_3 \times x_1 + w_4 x_2 + b_1$

$= 0.25 \times 0.05 + 0.3 \times 0.1 + 0.35$

$= 0.3925$

$out_h_2 = \frac{1}{1+e^{-h_2}} = \frac{1}{1+e^{-0.3925}} = 0.5969$

4) $net_O_1 = out_h_1 w_5 + out_h_2 w_6 + b_2$

$net_O_1 = 0.5933 * 0.4 + 0.5969 \times 0.45 + 0.6 = 1.105925$

$out_O_1 = \frac{1}{1+e^{-O_1}} = \frac{1}{1+e^{-1.105925}} = 0.7514$

$net_O_2 = out_h_1 w_7 + out_h_2 w_8 + b_2$

$= 0.5933 * 0.5 + 0.5969 \times 0.55 + 0.6$

$= 1.2249$

$out_O_2 = \frac{1}{1+e^{-O_2}} = \frac{1}{1+e^{-1.2249}} = 0.7729$

5) $E = \frac{1}{2}(0.01 - 0.7514)^2 + \frac{1}{2}(0.99 - 0.7729)^2$

$E = 0.2984$

6) we adjust weights using:-

$\frac{\partial E}{\partial w}$

$\frac{\partial E}{\partial w_5} = \frac{\partial E_{total}}{\partial out_{O1}} * \frac{\partial out_{O1}}{\partial net_{O1}} * \frac{\partial net_{O1}}{\partial w_5}$

$\frac{\partial E}{\partial O1} = \frac{\partial}{\partial out_{O1}} \frac{1}{2}(desired_1 - out_{O1})^2 + \frac{1}{2}(desired_2 - out_O_2)^2$

$= \cancel{2} \times \frac{1}{\cancel{2}}(desired_1 - out_O_1) \times (-1) + Zero$

$= out_O1 - desired_1 = \boxed{0.7514}$

$\frac{\partial out_{O1}}{\partial net_O1} = \frac{\partial}{\partial net_O1}\left(\frac{1}{1+e^{-net_{O1}}}\right) = \frac{1}{1+e^{-net_O1}}\left(1 - \frac{1}{1+e^{net_O1}}\right)$

$\boxed{= 0.1868}$

$\frac{\partial net_O1}{\partial w_5} = \frac{\partial}{\partial w_5}(out h_1 \times w_5 + out_h_2 w_6 + b_2)$

$= out_h_1 = \boxed{0.5933}$

$$\frac{\partial E}{\partial w_5} = 0.7414 \times 0.1868 \times 0.5933 = 0.08216$$

$$\boxed{w_{5new} = 0.4 - 0.5 \times 0.08216 = 0.3589}$$

$$\frac{\partial E_{total}}{\partial w_6} = \frac{\partial E_{total}}{\partial out\text{-}o_1} \times \frac{\partial out_{o1}}{\partial net\text{-}o_1} \times \frac{\partial net\text{-}o_1}{\partial w_6}$$

$$\frac{\partial E_{total}}{\partial out_{o1}} = 0.7414 \qquad \frac{\partial out_{o1}}{\partial net\text{-}o_1} = 0.1868$$

$$\frac{\partial net_{o1}}{\partial w_6} = \frac{\partial}{\partial w_6}\left(\cancel{out\text{-}h_1 w_5} + out\text{-}h_2 w_6 \cancel{+ b_2}\right)$$

$$= out\text{-}h_2 = 0.5969$$

$$\frac{\partial E}{\partial w_6} = 0.7414 \times 0.1868 \times 0.5969 = 0.082666$$

$$\boxed{w_{6\,new} = 0.45 - 0.5 \times 0.082666 = 0.408}$$

$$\frac{\partial E_{total}}{\partial w_7} = \frac{\partial E}{\partial out\text{-}o_2} \times \frac{\partial out\text{-}o_2}{\partial net\text{-}o_2} \times \frac{\partial net\text{-}o_2}{\partial w_7}$$

$$\frac{\partial E}{\partial out\text{-}o_2} = \frac{\partial}{\partial out\text{-}o_2} \cancel{\frac{1}{2}(desired\text{-}1 - out_{o1})^2} + \frac{1}{2}(desired\text{-}2 - out_{o2})^2$$

$$= \cancel{2 \times \frac{1}{2}(desired\text{-}2 - out_{o2}) \times (-1)}$$

$$= -(0.99 - 0.7729) = \boxed{-0.2171}$$

$$\frac{\partial out\text{-}o_2}{\partial net\text{-}o_2} = \frac{\partial}{\partial net\text{-}o_2} \frac{1}{1+e^{-net\text{-}o_2}} = \frac{1}{1+e^{-net\text{-}o_2}}\left(1 - \frac{1}{1+e^{-net\text{-}o_2}}\right)$$

$$= 0.7729\,(1 - 0.7729) = \boxed{0.175525}$$

$$\frac{\partial net\text{-}o2}{\partial w_7} = \frac{\partial}{\partial w_7} out\,h_1 \times w_7 + \cancel{out\,h_2 \times w_8 + b_2}$$

$$= out\,h_1 = 0.5933$$

$$\frac{\partial E}{\partial w_7} = -0.2171 \times 0.175525 \times 0.5933$$

$$= -0.022608$$

$$\boxed{w_{7new} = 0.5 - 0.5 * (-0.022608) = 0.5113}$$

$$\frac{\partial E}{\partial w_8} = \frac{\partial E}{\partial out_{o2}} * \frac{\partial out_{o2}}{\partial net_{o2}} * \frac{\partial net_{o2}}{\partial w_8}$$

$$\frac{\partial E}{\partial out_{o2}} = -0.2171 \qquad \frac{\partial out_{o2}}{\partial net_{o2}} = 0.175525$$

$$\frac{\partial net_{o2}}{\partial w_8} = \frac{\partial}{\partial w_8} \cancel{out\,h_1 \times w_7} + out\,h_2 \times \cancel{w_8 + b_2}$$

$$= out\,h_2 = 0.5969$$

$$\frac{\partial E}{\partial w_8} = -0.2171 \times 0.175525 \times 0.5969 =$$

$$= -0.02275$$

$$\boxed{w_{8new} = 0.55 - 0.5 \times (-0.02275) = 0.561375}$$

$$\frac{\partial E}{\partial out\,h_1} = \left(\frac{\partial E}{\partial out\text{-}o1} * \frac{\partial out_{o1}}{\partial net\,o1} * w_5\right) + \left(\frac{\partial E}{\partial out_{o2}} * \frac{\partial out_{o2}}{\partial net\,o2} * w_7\right)$$

$$= 0.7414 * 0.1868 \times 0.4 + (-0.2171 * 0.175525 * 0.5)$$

$$= 0.036344$$

$$\frac{\partial out\text{-}h_1}{\partial net\,h_1} = out\text{-}h_1 * (1 - out\,h_1)$$

$$= 0.5933 * (1 - 0.5933) = 0.2413$$

$$\frac{\partial net\,h_1}{\partial w_1} = x_1 = 0.05$$

$$\frac{\partial E}{\partial w_1} = 0.036344 * 0.2413 * 0.05 = 0.000438$$

$$\boxed{w_{1new} = 0.15 - 0.5 \times 0.000438 = 0.149781}$$

$$\frac{\partial out_{h1}}{\partial net_{h1}} = 0.2413 \qquad \frac{\partial net_{h1}}{\partial w_3} = x_2 = 0.1$$

$$\frac{\partial E}{\partial w_3} = 0.036344 * 0.2413 \times 0.1 = 0.000877$$

$$\boxed{w_{3new} = 0.25 - 0.5 \times 0.000877 = 0.2495615}$$

$$\frac{\partial E}{\partial out_{h2}} = \left(\frac{\partial E}{\partial out_{o1}} * \frac{\partial out_{o1}}{\partial net_{o1}} * w_6\right) + \left(\frac{\partial E}{\partial out_{o2}} * \frac{\partial out_{o2}}{\partial net_{o2}} * w_8\right)$$

$$= 0.7414 * 0.1868 * 0.45 = 0.062322$$

$$-0.2171 * 0.175525 * 0.55 = -0.0209$$

$$\frac{\partial E}{\partial out_{h2}} = 0.062322 + (-0.0209) = 0.041422$$

$$\frac{\partial out_{h2}}{\partial net_{h2}} = out_{h2}(1 - out_{h2})$$

$$= 0.5969(1 - 0.5969)$$

$$= 0.25690576$$

$$\therefore \frac{\partial E}{\partial w_2} = 0.041422 * 0.25690576 * 0.05 = 0.000532$$

$$\boxed{w_{2new} = 0.2 - 0.5 \times 0.000532 = 0.199734}$$

$$\frac{\partial out\text{-}h_2}{\partial net\,h_2} = 0.25690576 \qquad \frac{\partial net\,h_2}{\partial w_4} = x_2 = 0.1$$

$$\frac{\partial E}{\partial w_4} = 0.041422 \times 0.25690576 \times 0.1 = 0.0001$$

$$\boxed{w_{4\,new} = 0.3 - 0.5 \times 0.0001 = 0.2999}$$